

Ans.

|   |   |   |          |          |   |          |                   |                           |
|---|---|---|----------|----------|---|----------|-------------------|---------------------------|
| 1 |   |   |          |          |   |          | <u>1</u>          | ← Row 1, space = <u>6</u> |
| 1 | 2 |   |          |          |   |          | <u>2</u> <u>1</u> | ← Row 2, space = <u>4</u> |
| 1 | 2 | 3 |          |          |   | <u>3</u> | 2 <u>1</u>        | ← Row 3, space = <u>2</u> |
| 1 | 2 | 3 | <u>4</u> | <u>4</u> | 3 | 2        | <u>1</u>          | ← Row 4, space = <u>0</u> |

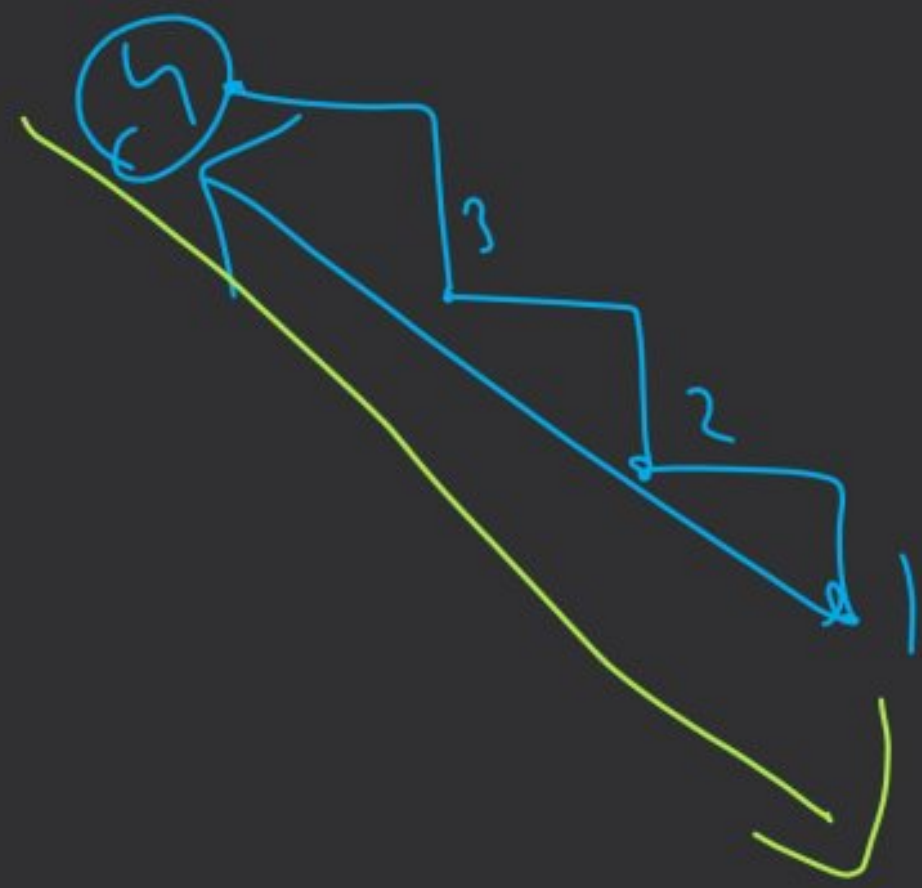
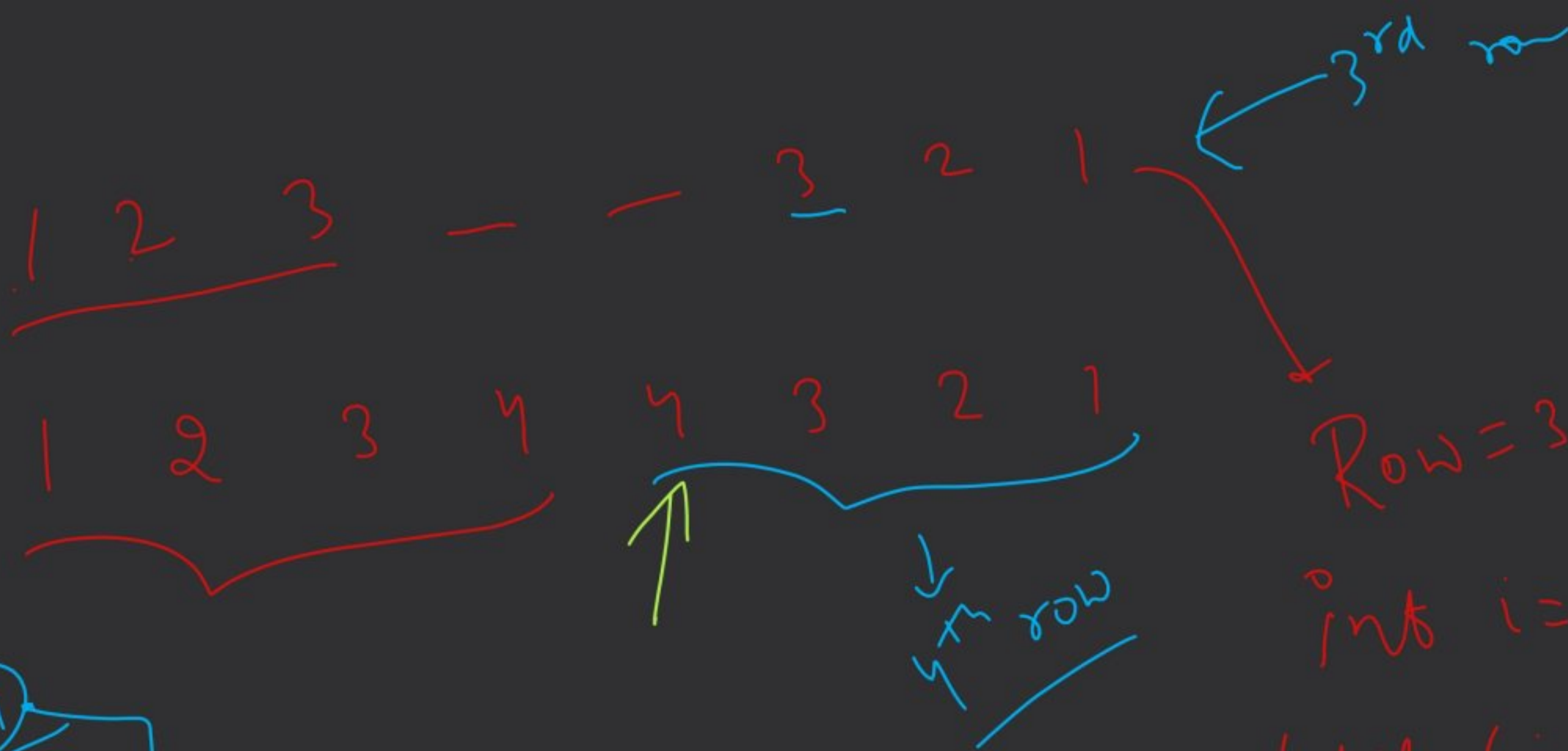
row

// left part

// spaces

// right part

(space -= 2)  
row++;



$k > 0$

Decreasing

Row = 3

```

int i = 1;
while (i <= 3) {
    Sout(i);
    i++;
}

```

```

// space
int j = 1;
while (j <= space) {
    Sout(" ");
    j++;
}

int k = row;
while (k >= 1) {
    Sout(k);
    k--;
}

```



```
int n=4, row=1, space=6;
```

```
while (row <= n) {
```

```
// Left portion
```

```
int i=1;
```

```
while (i <= row) {
```

```
    Sout(i);
```

```
    i++;
```

```
}
```

```
// space
```

```
int j=1;
```

```
while (j <= space) {
```

```
    Sout(" ");
```

```
    j++;
```

```
}
```

```
// Right portion
```

```
int k=row;
```

```
while (k >= 1) {
```

```
    Sout(k);
```

```
    k--;
```

```
    Sout("\n");
```

```
    row++;
```

```
}
```

```
✓ α 4
```

```
✓ α 4
```

```
✓ α 4
```

|                     |                            |                            |                            |                            |                |       |
|---------------------|----------------------------|----------------------------|----------------------------|----------------------------|----------------|-------|
| <u>(1,1)</u><br>1 ✓ |                            |                            |                            |                            |                | Row 1 |
| (2,1)<br>2 ✓        | <u>(2,2)</u><br><u>2 ✓</u> |                            |                            |                            |                | 2     |
| (3,1)<br>3 ✓        | (3,2)<br>0                 | <u>(3,3)</u><br><u>3 ✓</u> |                            |                            |                | 3     |
| (4,1)<br>4 ✓        | (4,2)<br>0                 | (4,3)<br>0                 | <u>(4,4)</u><br><u>4 ✓</u> |                            |                | 4     |
| → (5,1)<br>5 ✓      | (5,2)<br>0                 | (5,3)<br>0                 | (5,4)<br>0                 | <u>(5,5)</u><br><u>5 ✓</u> |                | 5     |
| <u>6</u> ✓          | 0                          | 0                          | 0                          | 0                          | <u>6</u>       |       |
| (6,1)               | (6,2)                      | (6,3)                      | (6,4)                      | (6,5)                      | <u>(6,6)</u> ✓ |       |

$col = -1$   
 ↓  
 print(row)  
 F  
 if ( $col == -1$  ||  $col == row$ )

$a = 2$  ||  $n = 3$



```
if (col == 1 || col == row) {
```

```
    cout << " ";
```

```
}
```

```
else
```

```
    cout << "0";
```

↔  
Similar

```
if (col == 1)
```

```
    sort(row);
```

```
else if (col == row)
```

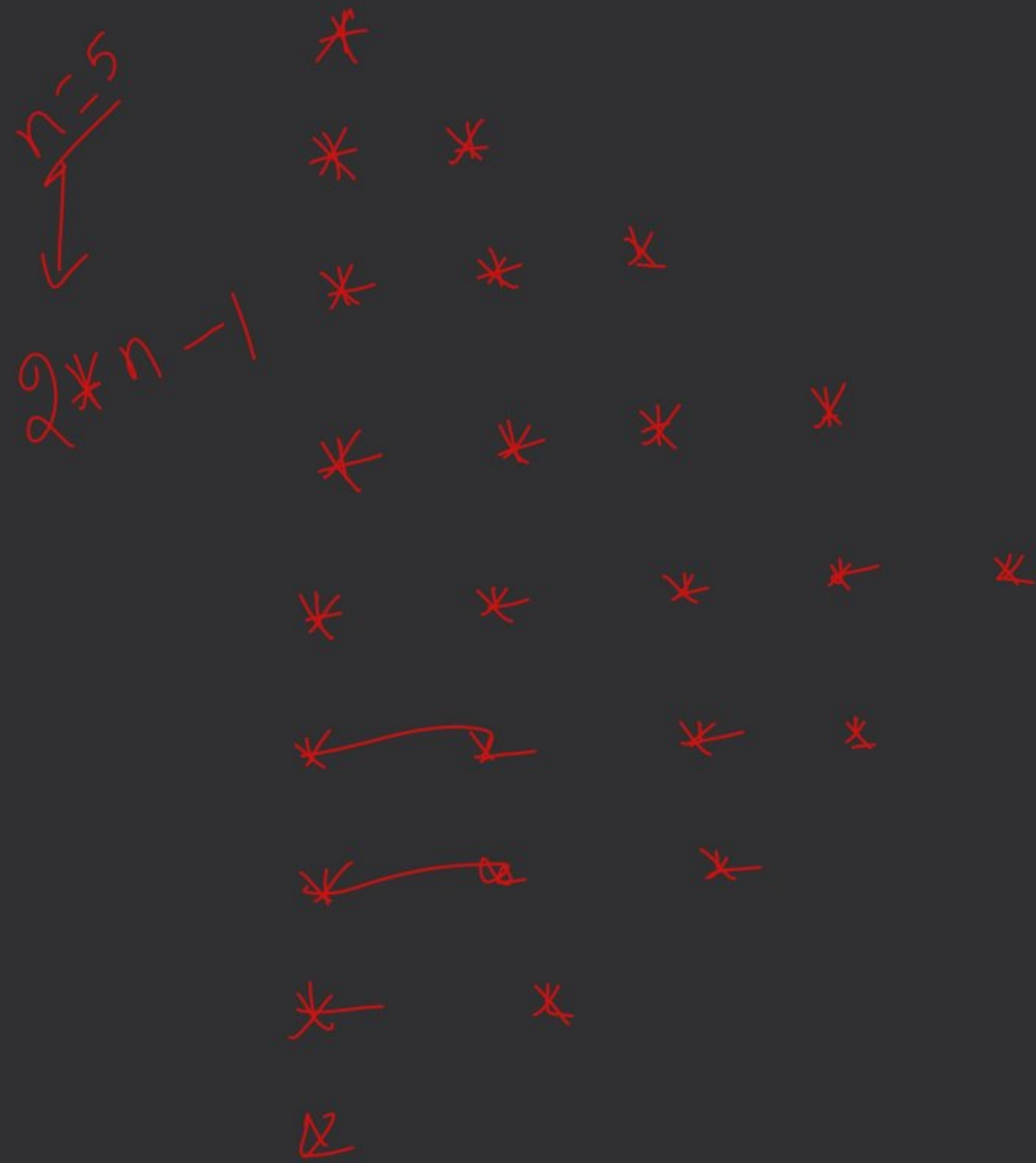
```
    sort(row);
```

```
else
```

```
    sort(0);
```

```
int row = 1; 2, 3  
while (row <= n) {
```

```
    int col = 1; ←  
    while (col <= 100) {  
        ...  
    }
```



$n=5$

as (1 to 9), star 21;

while (row <= 2 \* n - 1)

int i = 1;

while (i <= 8 + n)

cout << " \* ";

i++;

}

cout << endl;

row++;

Star++;



H.W.

n=5

-----\*

-----\*\*

-----\*\*\*

-----\*\*\*\*

\*\*\*\*\*

-----\*\*

-----\*\*\*

-----\*\*\*\*

Must Do for everyone

→ Do Dry Run on  $n=5$

→ Take different  $n$

→ Do the code





| row | space | star |
|-----|-------|------|
| 1   | 0     | 5    |
| 2   | 2     | 4    |
| 3   | 4     | 3    |
| 4   | 6     | 2    |
| 5   | 8     | 1    |
|     | 6     | 2    |

```

n = 5
(Star = n, Space = 0, row = 1)
while (row <= 2 * n - 1) {
  int i = 1;
  while (i <= space) {
    cout << " ";
    i++;
  }
  cout << "* ";
  row++;
}
  
```

```

// Star <
int j = 1;
while (j <= star) {
  cout << "* ";
  j++;
}
  
```

```

if (row < n) {
  Star--;
  Space = Space + 2;
} else {
  Star++;
  Space -= 2;
  cout << "\n";
  row++;
}
  
```



N=7

Ques.   
 \* \* \* \_ \* \* \*   
 \* \* \_ \_ \_ \* \*   
 \* \_ \_ \_ \_ \_ \*   
 \_ \_ \_ \_ \_   
 \* \_ \_ \_ \_ \*   
 \* \* \_ \_ \_ \* \*   
 \* \* \* \_ \* \* \*

row < n

Terminal

```

* * * _ * * *
* * _ _ _ * *
* _ _ _ _ *
_ _ _ _ _
* _ _ _ _ *
  
```

Row = 1, Star =  $\frac{n}{2}$  / Space = 1

// star

// space

// star

if (row <  $\frac{n}{2} + 1$ )

{ star--; space += 2;

else { star++;

space -= 2;

sample();

col++;

}

N=7, Row=X, Star=\*, Space=X

|   |   |   |
|---|---|---|
| 2 | 2 | 3 |
| 3 | 1 | 5 |
| 4 | 0 | 7 |
| 5 | 1 | 5 |
| 6 | 2 | 3 |
| 7 | 3 | 1 |





```

1  public class Pattern13 {
    Run | Debug
2  public static void main(String[] args) {
3      int n = 5;
4      int row = 1, star = 1;
5      *
6      * * while(row <= 2*n-1){
7      * * * int i = 1;
8      * * * while(i <= star){
9      * * * * System.out.print(s: "*");
10     * * * * i++;
11     * * * }
12
13     //mirror
14     if(row < n) star++;
15     else star--;
16
17     System.out.println();
18     row++;
19
20     // System.out.println(); //next line
21     // star++; //star is increasing
22     // row++; //row is increasing
23     // }
    }
}

```

| row | star |
|-----|------|
| 1   | 1    |
| 2   | 2    |
| 3   | 3    |
| 4   | 4    |
| 5   | 5    |
| 6   | 4    |
|     | 3    |
|     | 2    |
|     | 1    |



```

1  public class Pattern13 {
    Run | Debug
2  public static void main(String[] args) {
3      int n = 5;
4      int row = 1, star = 1;
5
6      while(row <= 2*n-1){
7          int i = 1;
8          while(i <= star){
9              System.out.print(s: "*");
10             i++;
11         }
12
13         //mirror
14         if(row < n) star++;
15         else star--;
16
17         System.out.println();
18         row++;
19
20         // System.out.println(); //next line
21         // star++; //star is increasing
22         // row++; //row is increasing
23         // }
    }
}

```

| row | star |
|-----|------|
| 1   | 1    |
| 2   | 2    |
| 3   | 3    |
| 4   | 4    |
| 5   | 5    |
| 6   | 4    |
|     | 3    |
|     | 2    |
|     | 1    |



```
File Edit Selection View Go Run Terminal Help
Pattern1.java Pattern2.java Pattern3.java Pattern4.java Divisors.java Pattern13.java Pattern12.java Pattern14.java Pattern15.java X
Pattern15.java > Pattern15 > main(String[])
1 public class Pattern15 {
2     public static void main(String[] args) {
3         int n = 5;
4         int space = 0;
5         int row = 1, star = n;
6
7         while(row <= 2*n-1){
8             //space
9             int i = 1;
10            while(i <= space){
11                System.out.print(s: " ");
12                i++;
13            }
14            //star
15            int j = 1;
16            while(j <= star){
17                System.out.print(s: "*");
18                j++;
19            }
20
21            if(row < n){
22                space += 2;
23                star -= 1;
24            }else{
25                space -= 2;
26                star += 1;
27            }
28
29            System.out.println();
30            row++;
31        }
    }
}
```

space = 0, 2, 4, 6, 4, 2, 0

I  
→ star = 5, 4, 3, 2, 1, 2, 3, 4, 5



```
Pattern1.class
Pattern1.java
Pattern2.class
Pattern2.java
Pattern3.class
Pattern3.java
Pattern4.java
Pattern12.class
Pattern12.java
Pattern13.class
Pattern13.java
Pattern14.class
Pattern14.java
Pattern15.class
Pattern15.java

public static void main(String[] args) {
    int n = 5;
    int space = 0;
    int row = 1, star = n;

    while(row <= 2*n-1){
        //space
        int i = 1;
        while(i <= space){
            System.out.print(s: " ");
            i++;
        }
        //star
        int j = 1;
        while(j <= star){
            System.out.print(s: "*");
            j++;
        }

        if(row < n){
            space += 2;
            star -= 1;
        }else{
            space -= 2;
            star += 1;
        }

        System.out.println();
        row++;
    }
}
```

Pattern13.java &gt; Pattern13 &gt; main(String[])

```
1  public class Pattern13 {
2      public static void main(String[] args) {
14         // if(row < n) star++;
15         // else star--;
16
17         // System.out.println();
18         // row++;
19
20         System.out.println(); //next line
21         star++; //star is increasing
22         row++; //row is increasing
23     }
24 }
25
26
```

| Row | Star |
|-----|------|
| 1   | 1    |
| 2   | 2    |
| 3   | 3    |
| 4   | 4    |
| 5   | 5    |
| 6   | 6    |
| 7   | 7    |
| 8   | 8    |
| 9   | 9    |

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Code + ▼ 📄 🗑️ ... 🔍 ✕

```
***
****
*****
*****
*****
*****
*****
*****
*****
```

PS C:\Users\91950\Desktop\coding blocks\lecture 9&gt; █